

CTA200 Assignment 3

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1 Question 1: The Mandelbrot Set

1.1 Methods

For each point $c = x + iy$ in the complex plane, with $-2 < x < 2$ and $-2 < y < 2$, we set $z_0 = 0$ and iteratively apply the equation:

$$z_{i+1} = z_i^2 + c \quad (1)$$

A point is considered to diverge if $|z|^2 > 4$ at any iteration. The iteration was performed up to a maximum of 100 steps. The function was implemented in a separate file `mandelbrot.py` and imported into the notebook.

1.2 Results

Figure 1 shows the Mandelbrot set, where bounded points are shown in red and diverging points in blue. Figure 2 shows the same set colored by the iteration number at which each point diverged, revealing the complex boundary structure of the set.

2 Question 2: The Lorenz System

2.1 Methods

The Lorenz equations are a system of three coupled ordinary differential equations:

$$\dot{X} = -\sigma(X - Y) \quad (2)$$

$$\dot{Y} = rX - Y - XZ \quad (3)$$

$$\dot{Z} = -bZ + XY \quad (4)$$

where the parameters used are $\sigma = 10$, $r = 28$, and $b = 8/3$, with initial conditions $W_0 = [0, 1, 0]$. The equations were solved using `solve_ivp` from `scipy.integrate` over $t = 60$ dimensionless time units with a maximum step size of $\Delta t = 0.01$.

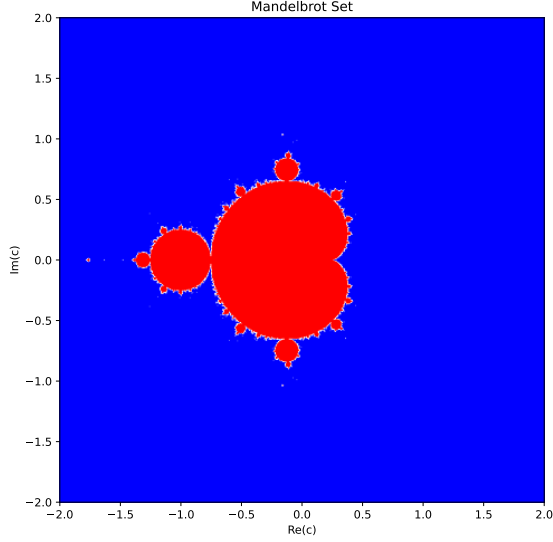


Figure 1: Mandelbrot set: bounded points vs diverging points.

2.2 Results

Figure 3 reproduces Lorenz's Figure 1, showing Y as a function of iteration number $N = t/\Delta t$ over 3000 iterations. The solution shows increasingly irregular oscillations over time.

Figure 4 reproduces Lorenz's Figure 2, showing projections of the solution trajectory in the Y - Z and X - Y planes between iterations 1400 and 1900. The characteristic butterfly-shaped attractor is visible.

To investigate sensitivity to initial conditions, the equations were solved again with a slightly perturbed initial condition $W'_0 = W_0 + [0, 10^{-8}, 0]$. The distance between the two solutions was calculated as:

$$d(t) = \sqrt{(X - X')^2 + (Y - Y')^2 + (Z - Z')^2} \quad (5)$$

Figure 5 shows this distance on a semilog plot. The straight line up to $t \approx 30$ confirms exponential growth of the initial error, demonstrating the chaotic nature of the Lorenz system. After $t \approx 30$, the distance saturates as the solutions become completely independent.

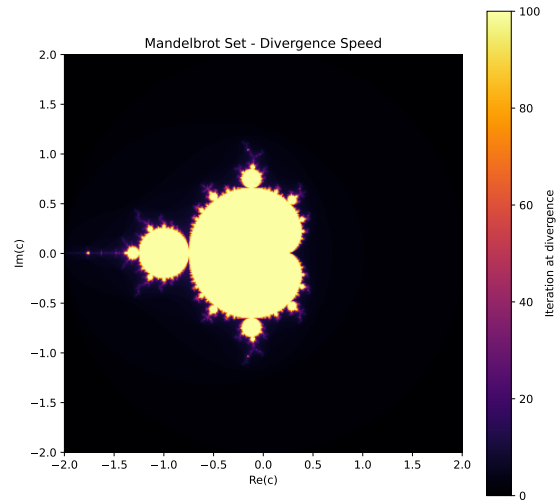


Figure 2: Mandelbrot set colored by divergence iteration number.

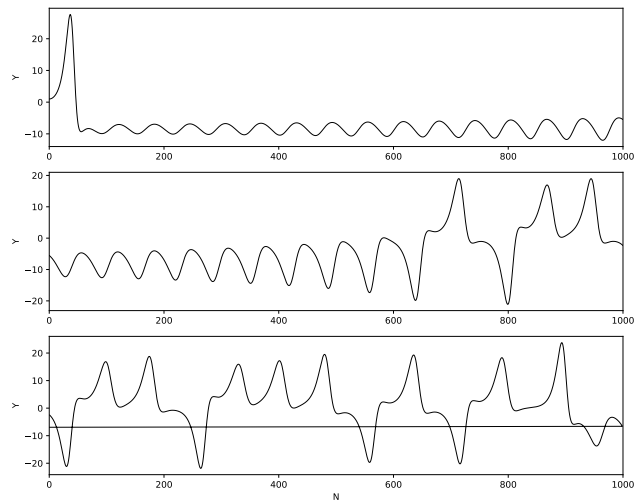


Figure 3: Reproduction of Lorenz Figure 1: Y vs N over 3000 iterations.

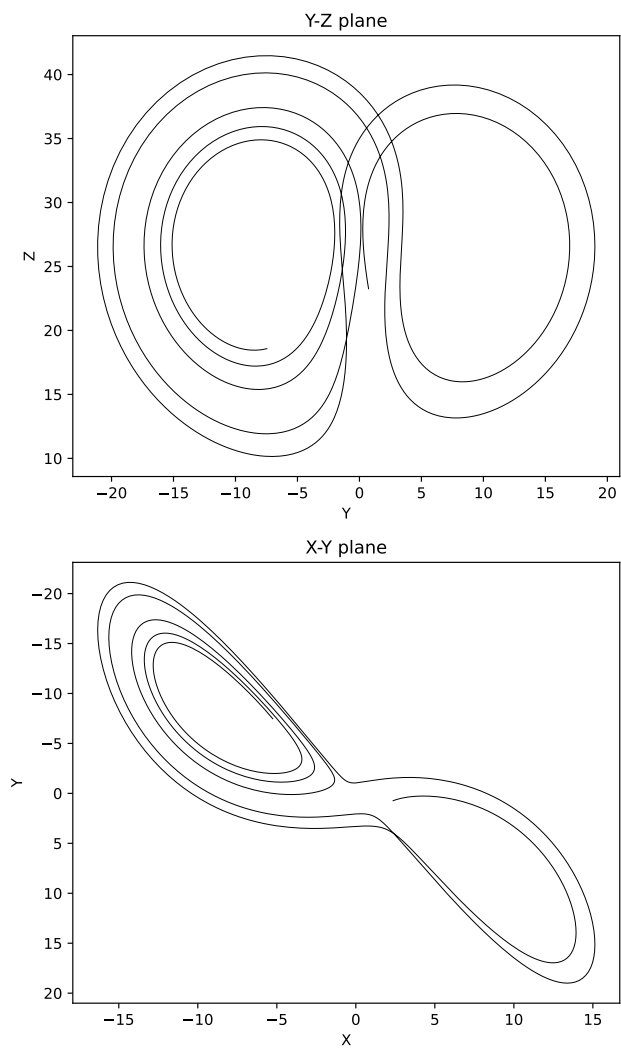


Figure 4: Reproduction of Lorenz Figure 2: phase space projections.

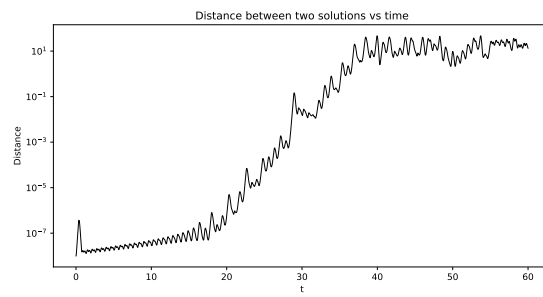


Figure 5: Distance between two solutions with slightly different initial conditions on a semilog plot.